

Deployment of a Machine Learning / Time Series model.

What is Deployment?

Deployment means making your trained model available for real-world use.

Example:

Train Model in Google Colab



Save Model



Deploy Model



Users Can Use It

Step 1: Load the Saved Model

Load the saved model

After training, you save the model:

```
model.save("stock_prediction_model.h5")
```

Later, load it:

```
from tensorflow.keras.models import load_model
```

```
model = load_model("stock_prediction_model.h5")
```

Example

You trained a model to predict tomorrow's stock price.

Now you load that saved model instead of training again.

Step 2: Get Inputs

Get inputs

Collect data from the user.

Example

User enters:

Input	Value
Today's Price	₹100
Yesterday's Price	₹95
Volume	10,000

These become model inputs.

Step 3: Predicts

Predicts

The model processes the input and generates an output.

Example

Input:

Today's Price = ₹100

Model Prediction:

Tomorrow = ₹105

or

Prediction:

BUY

Step 4: Call to Action

Call to action

Take action based on the prediction.

Example

Prediction	Action
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Price ↑	Buy
Price ↓	Sell
Waste = Plastic	Put in Plastic Bin
Dog Breed = Labrador	Display Result

Complete Flow

Model Training
↓
Save Model
↓
Deployment
↓
Load Model
↓
Get User Input
↓
Predict
↓
Show Result / Take Action

Real Example (Your Waste Classification Project)

Train VGG16 Model
↓
Save waste_vgg16.h5
↓
Deploy in Streamlit
↓
User Uploads Image
↓
Model Predicts
↓
Plastic / Paper / Glass
↓

Display Result

Real Example (Stock Market)

Historical Stock Data



Train Model



Save Model



User Enters Today's Price



Model Predicts Tomorrow's Price



Buy / Sell Decision

Deployment is the process of loading a trained model, accepting user input, generating predictions, and providing results or actions in a real-world application.

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